

Goal

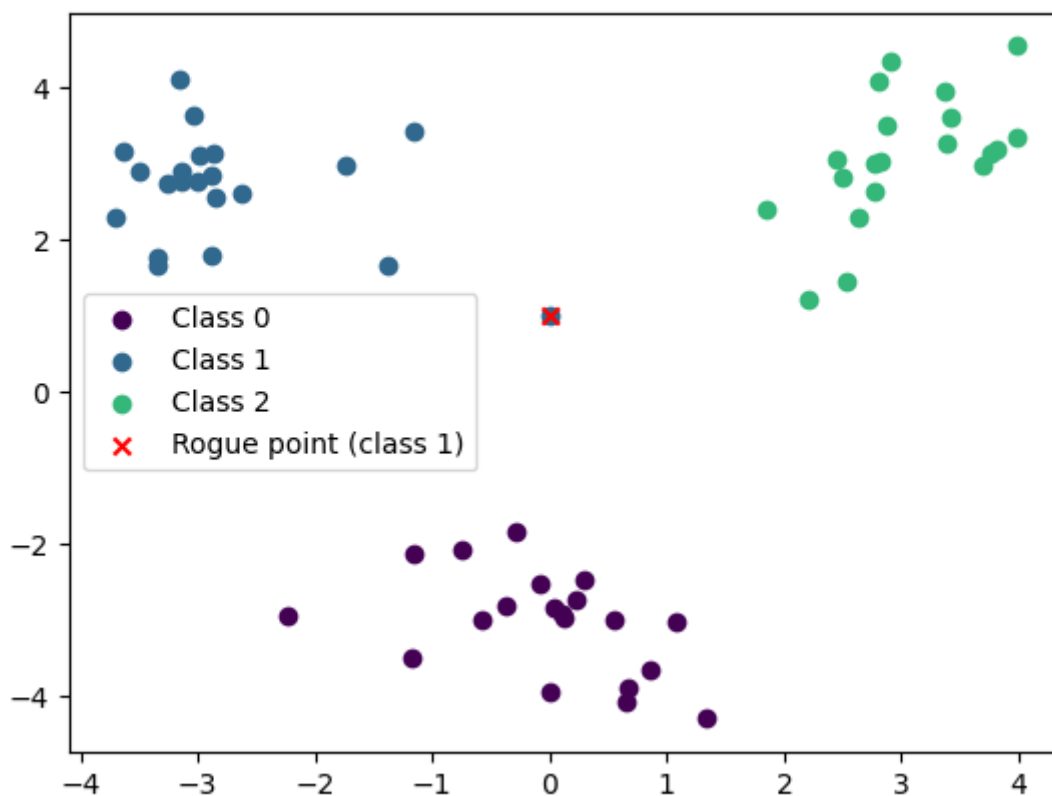
In this experiment we want to investigate the behavior of the SSD method and how the choice of hyperparameters affect the decision boundary. We also try to address the shortcomings of SSD in two ways:

- By applying dampening at only the last layer of the model.
- By searching for hyperparameters whose dampening yields a loss as close as possible to the original model's loss on a "generalization dataset" constructed in the same way as the repair step for Scrub+R.

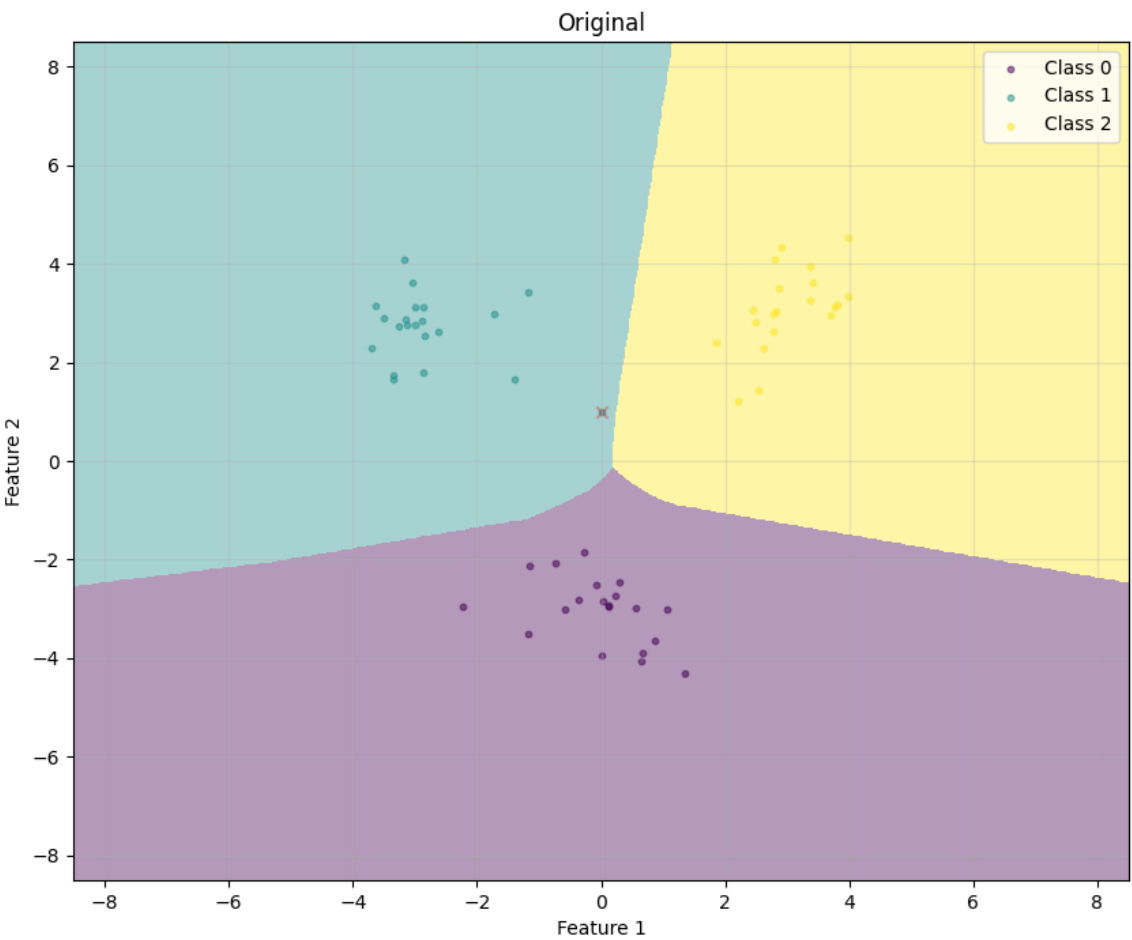
Experiment runs

Now we run the experiment on each of the dataset configuration, moving the rogue point.

1. Rogue point with same distance to all centroids (close to decision boundary)



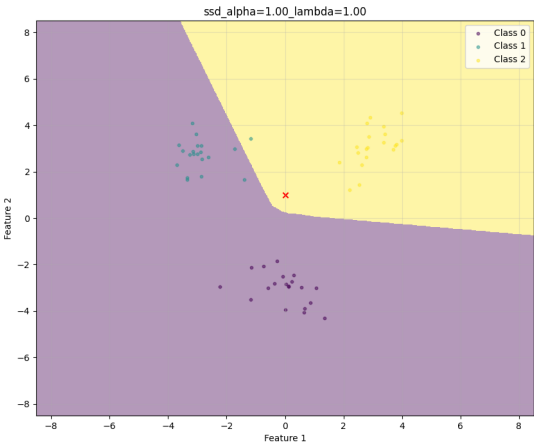
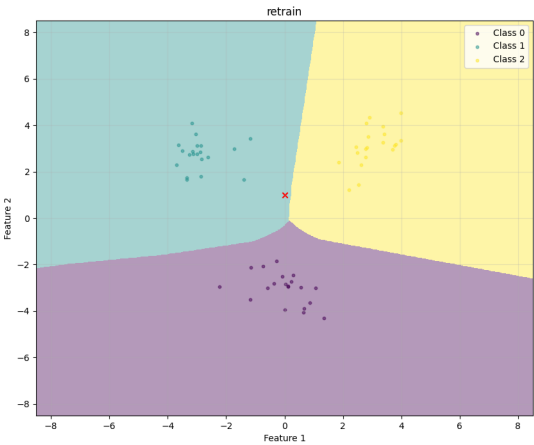
Original model for the dataset

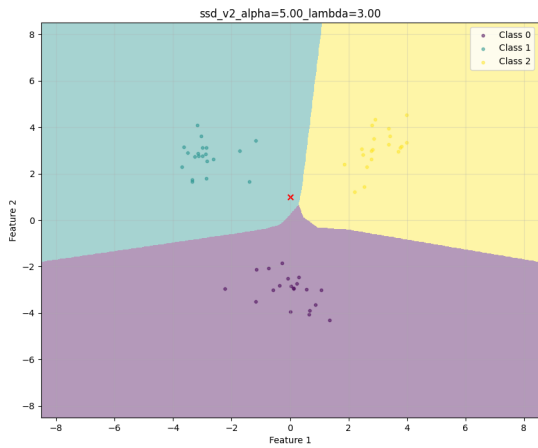
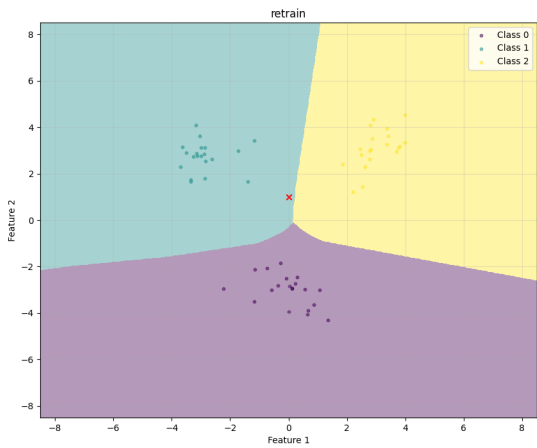
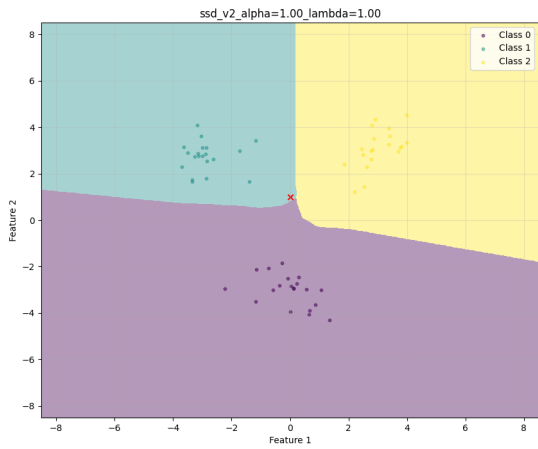
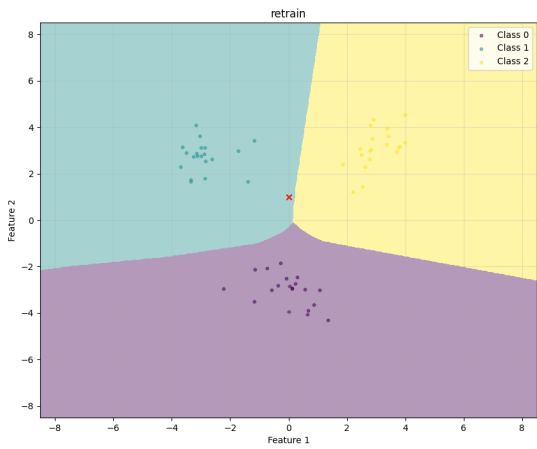
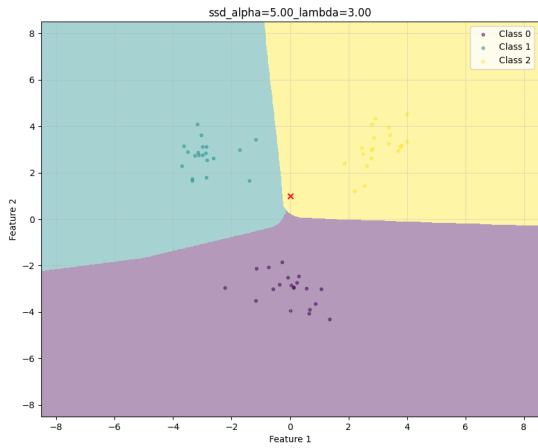
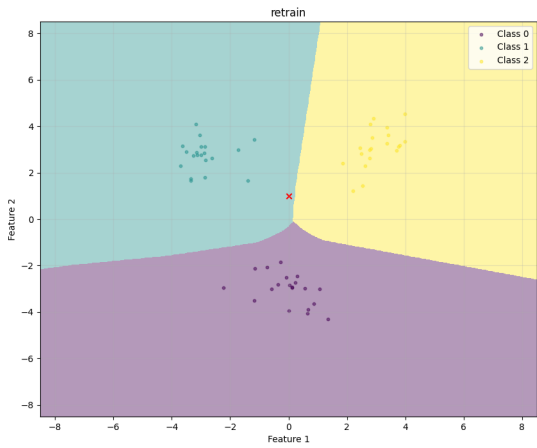


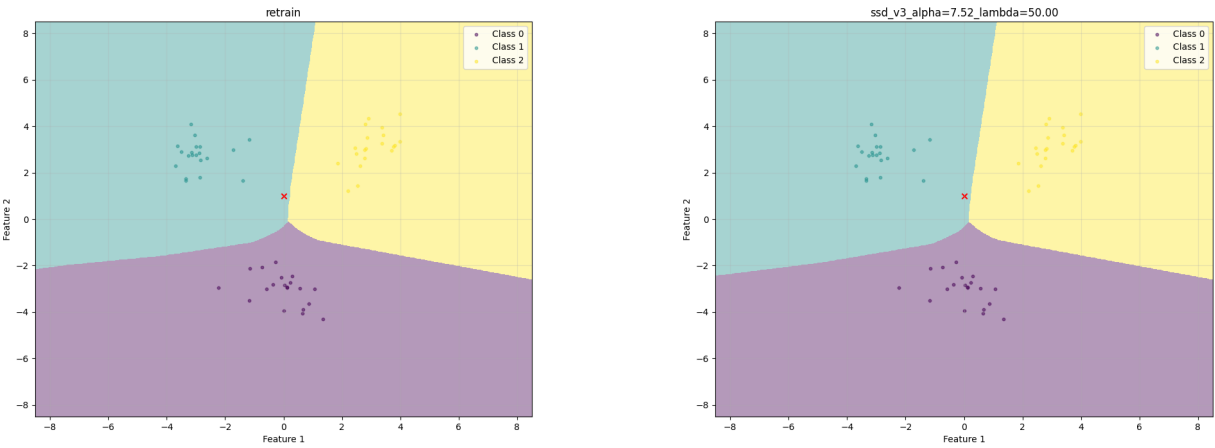
Findings

Unlearned

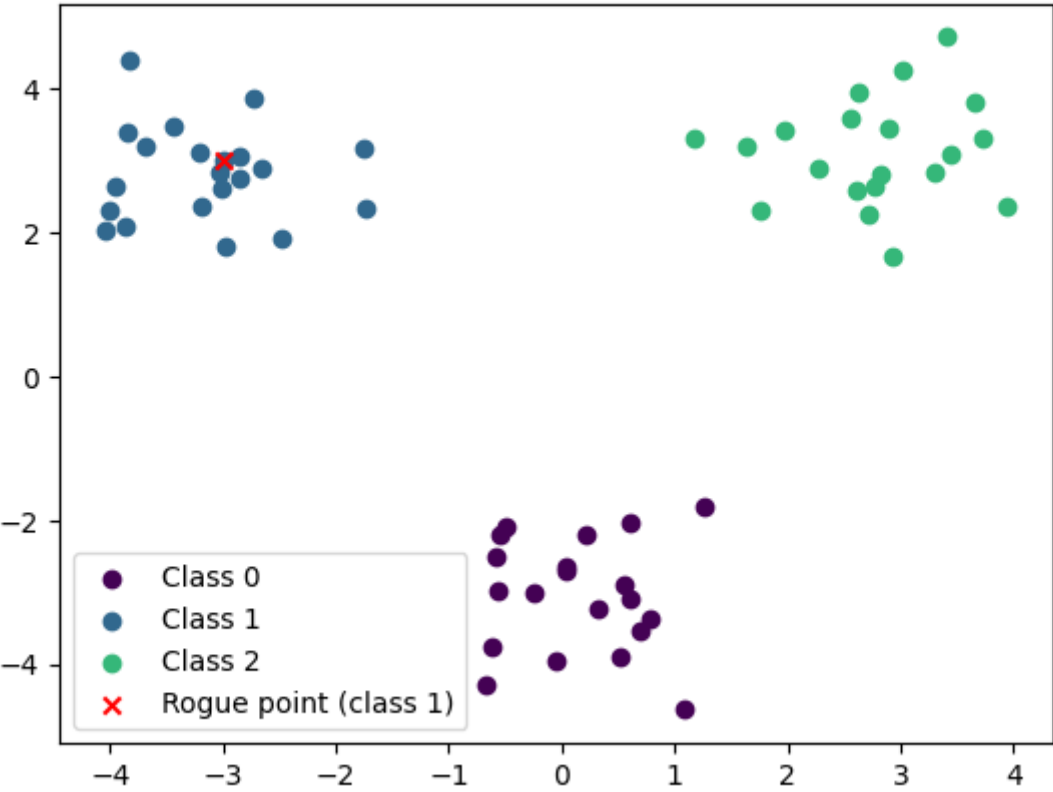
SSD



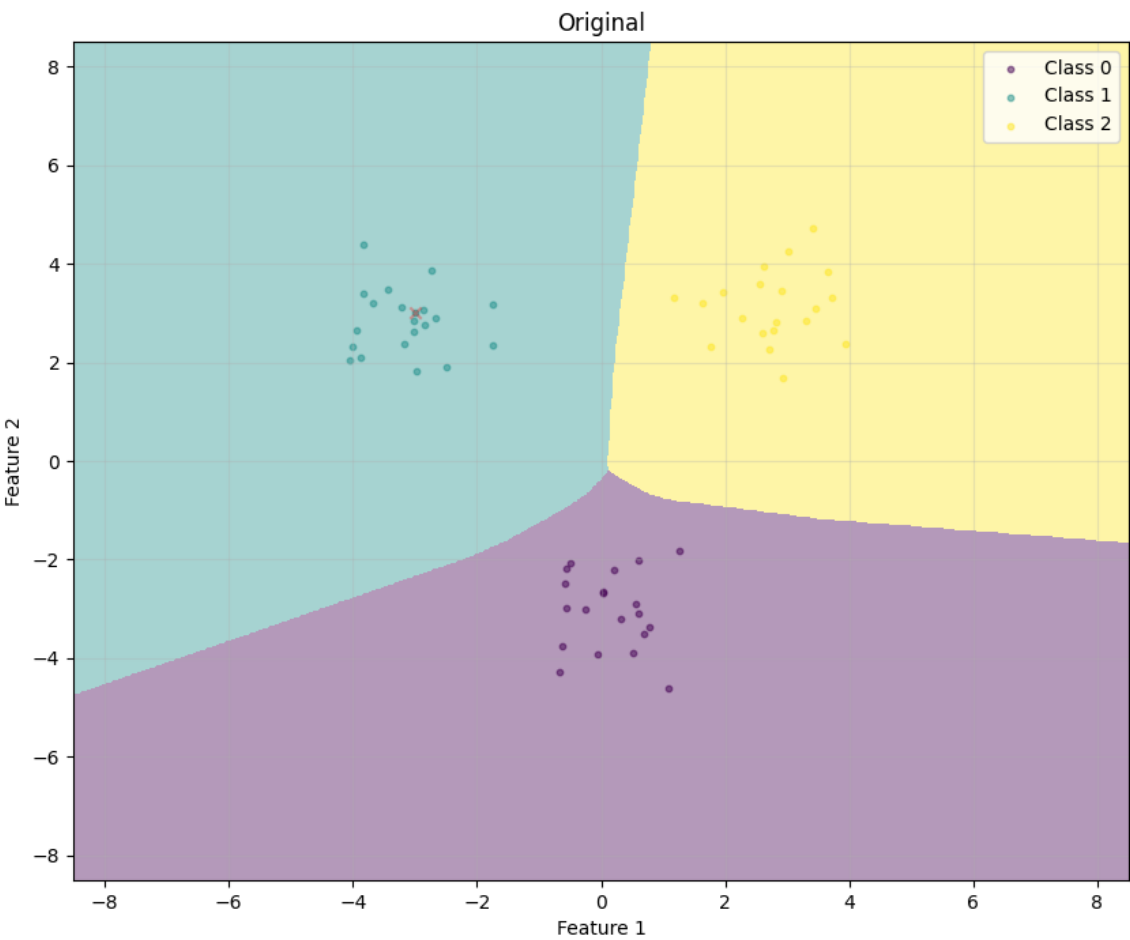




2. Rogue point with same centroid as its class

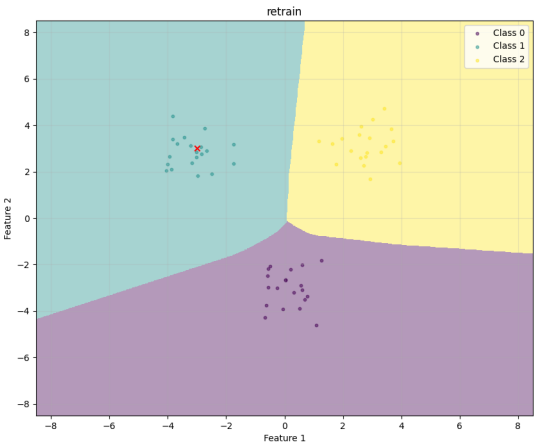


Original model for the dataset

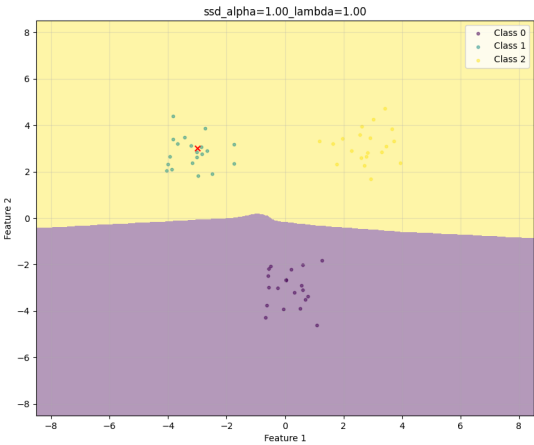


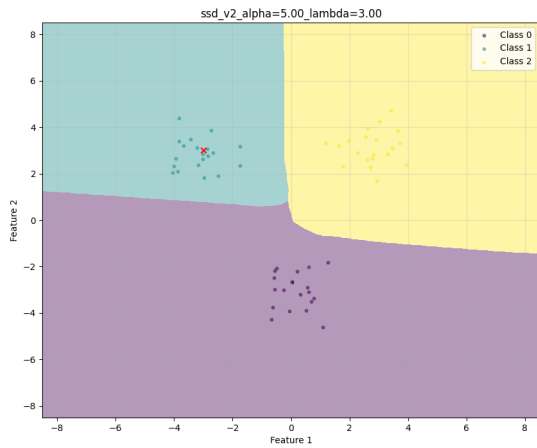
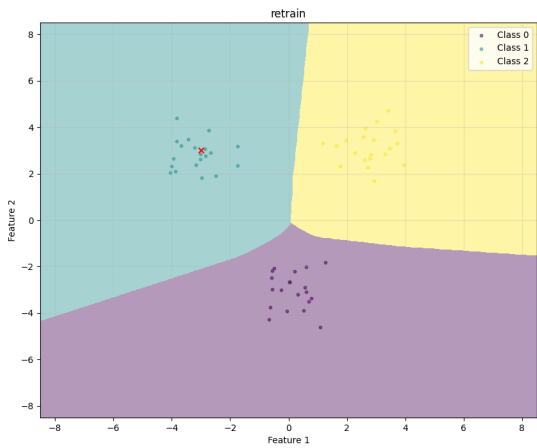
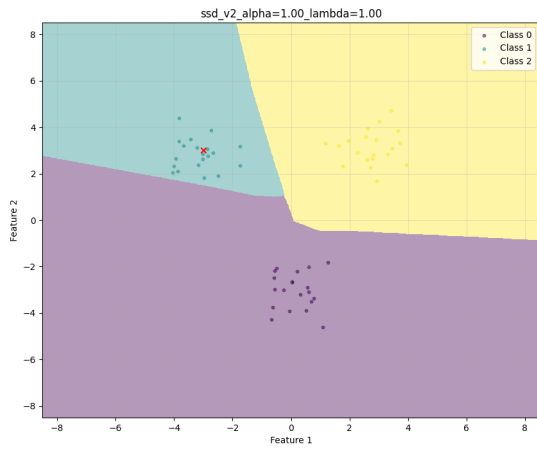
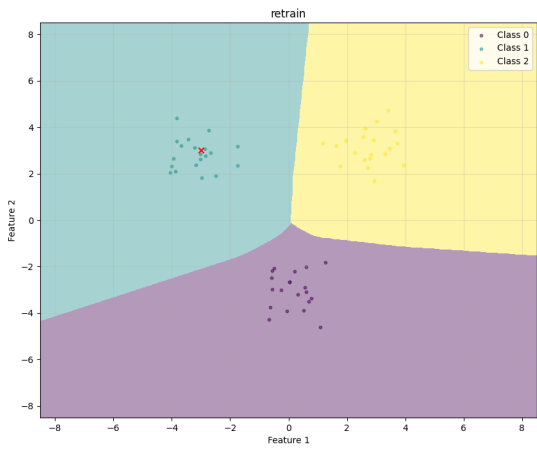
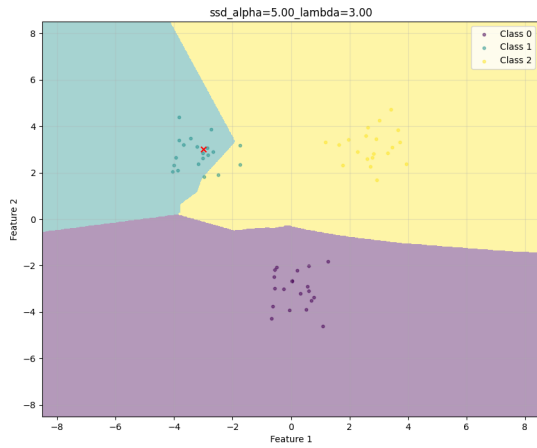
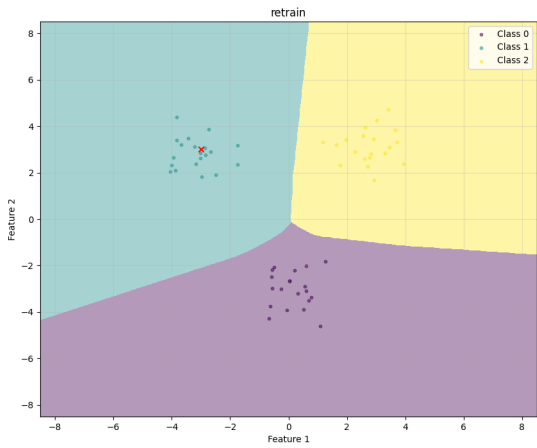
Findings

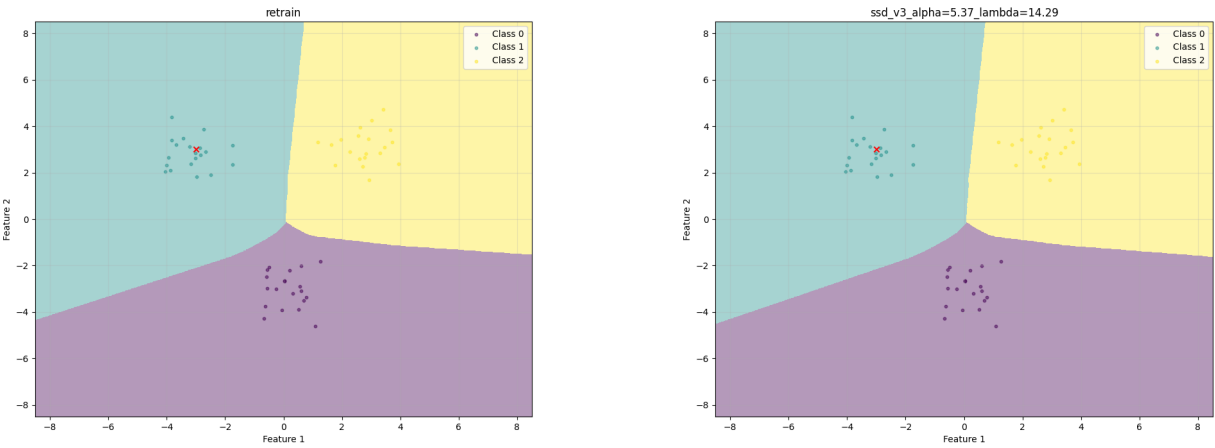
Unlearned



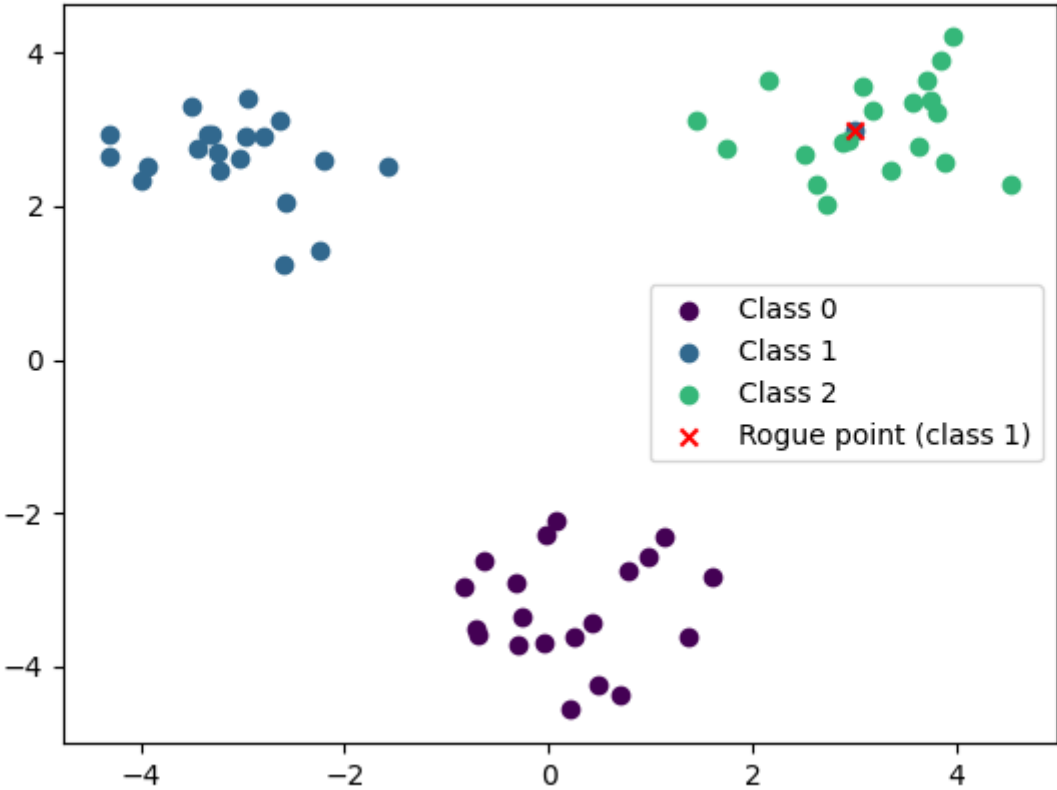
SSD



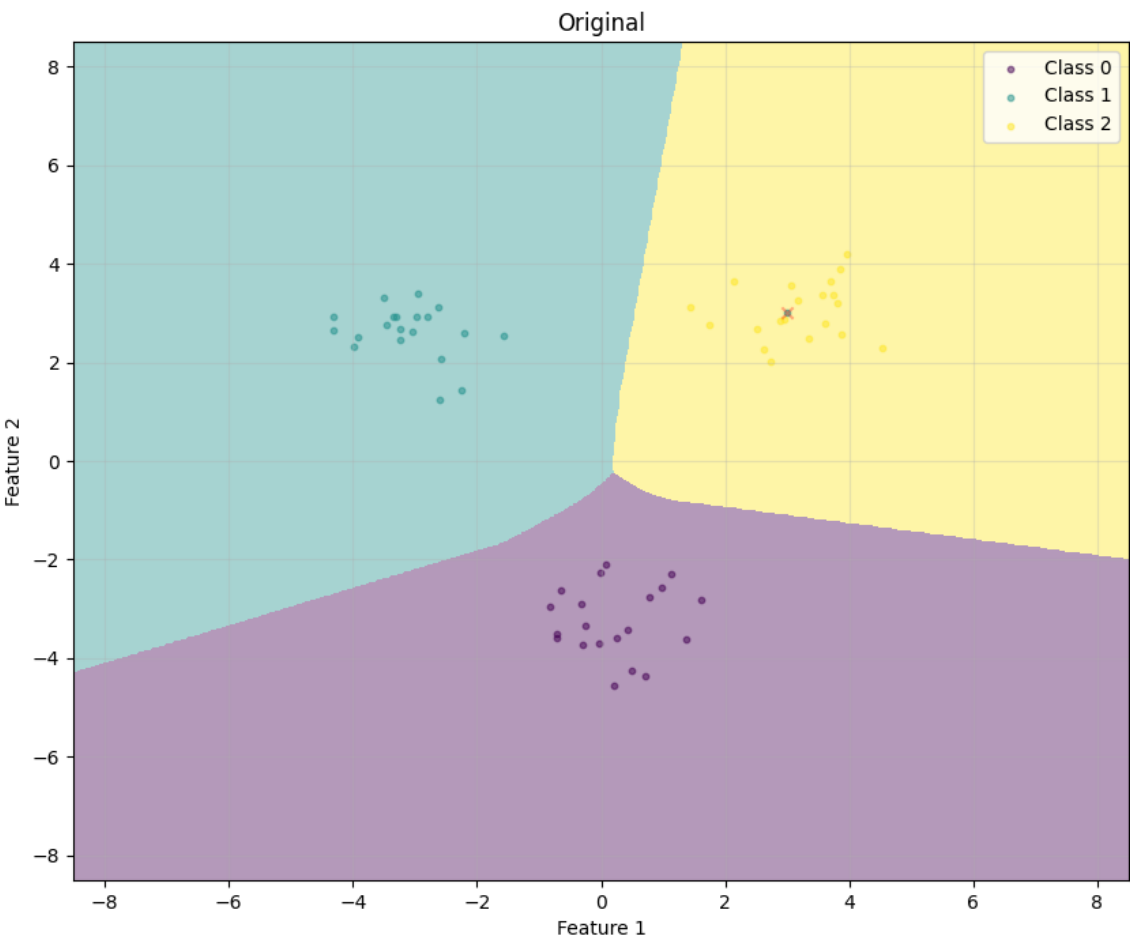




3. Rogue point with same centroid as the class with a different label



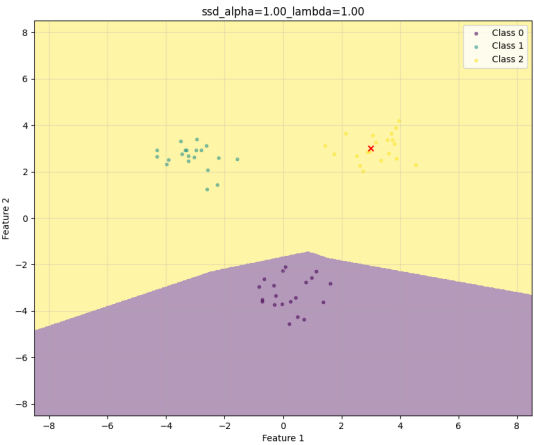
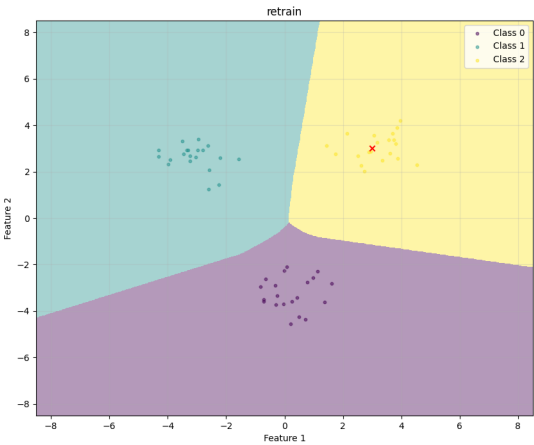
Original model for the dataset

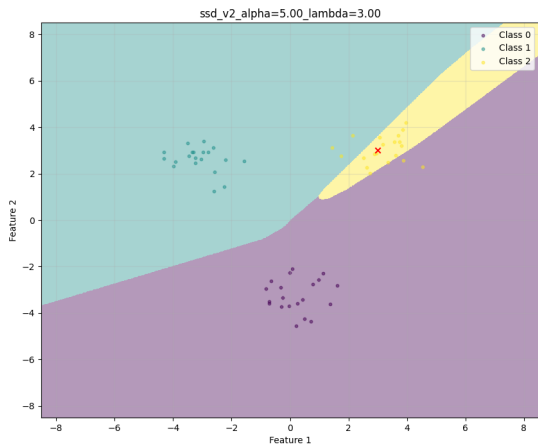
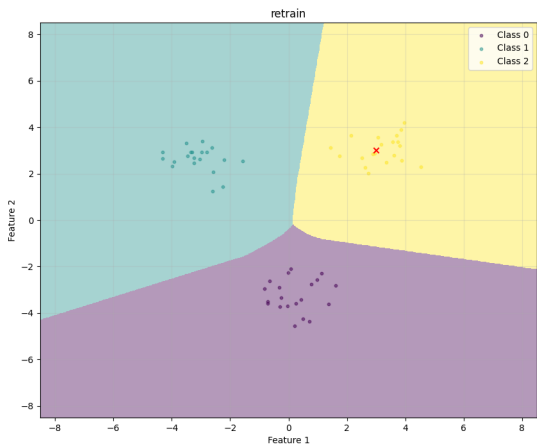
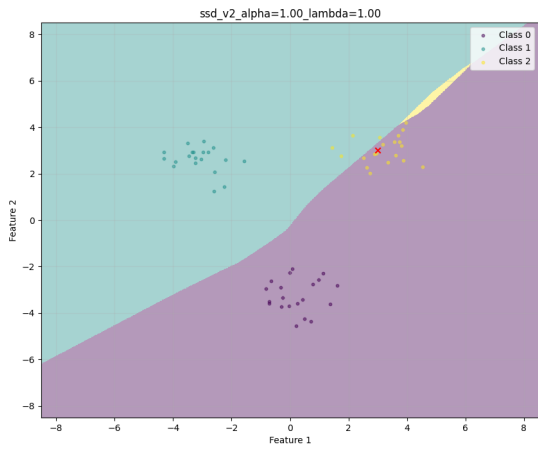
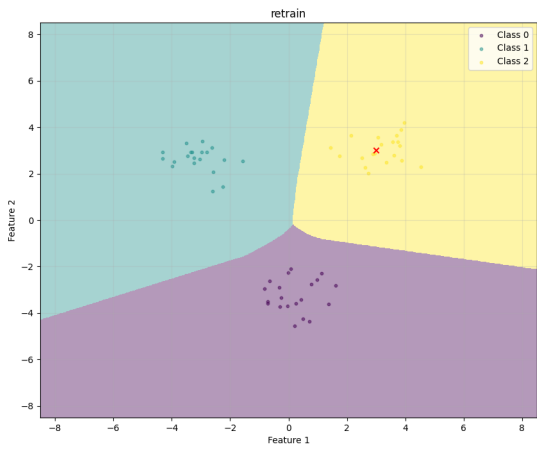
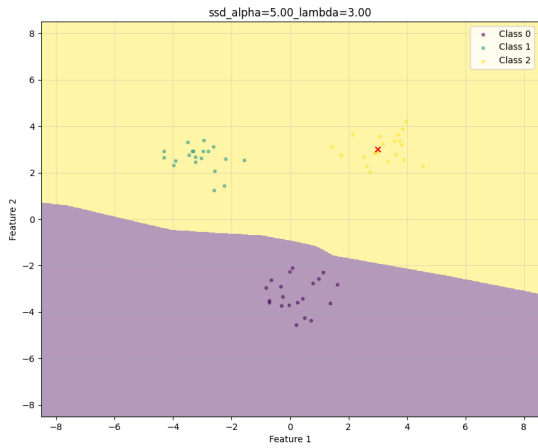
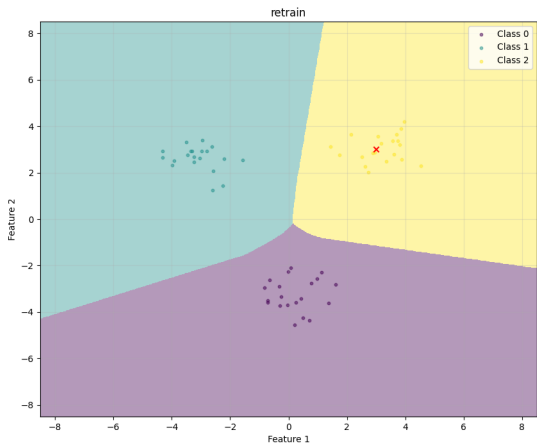


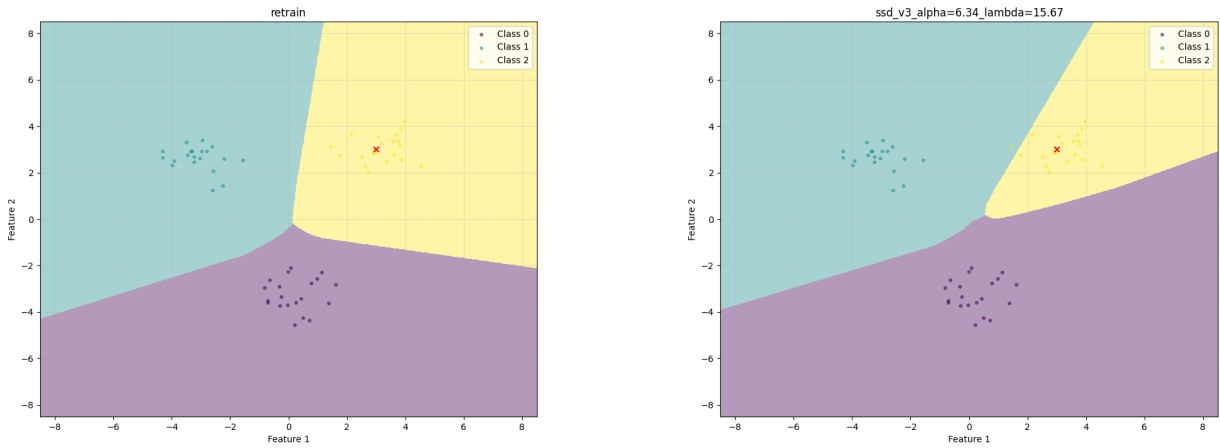
Findings

Unlearned

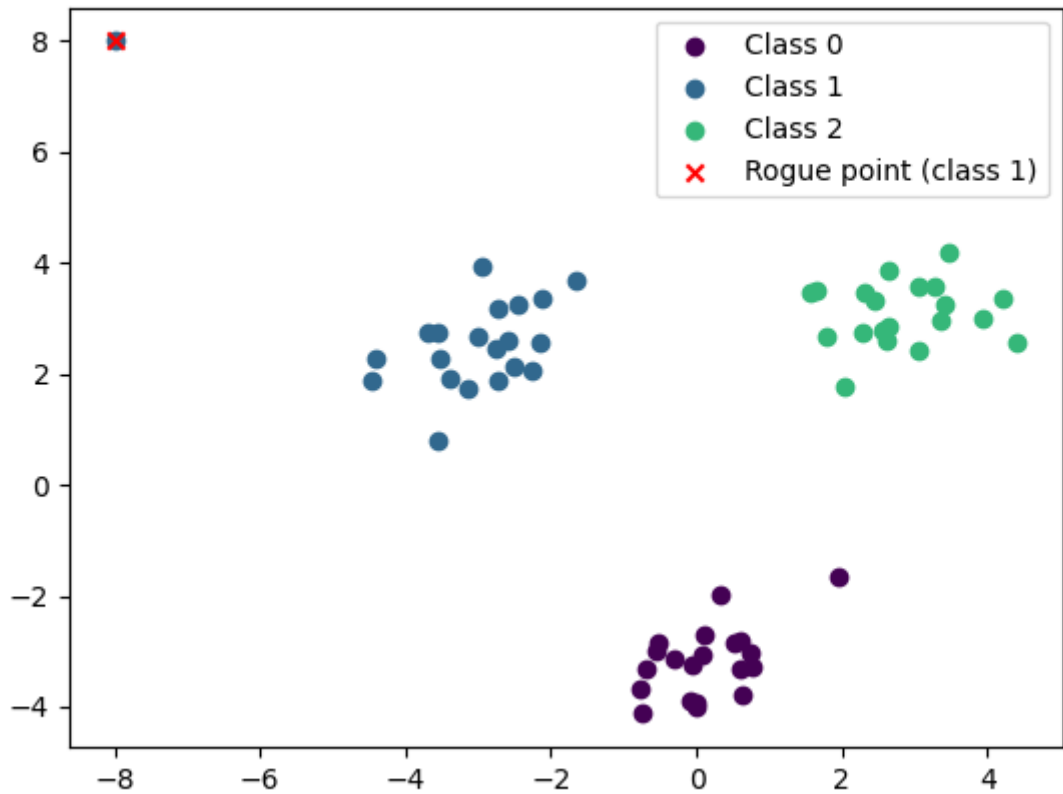
SSD



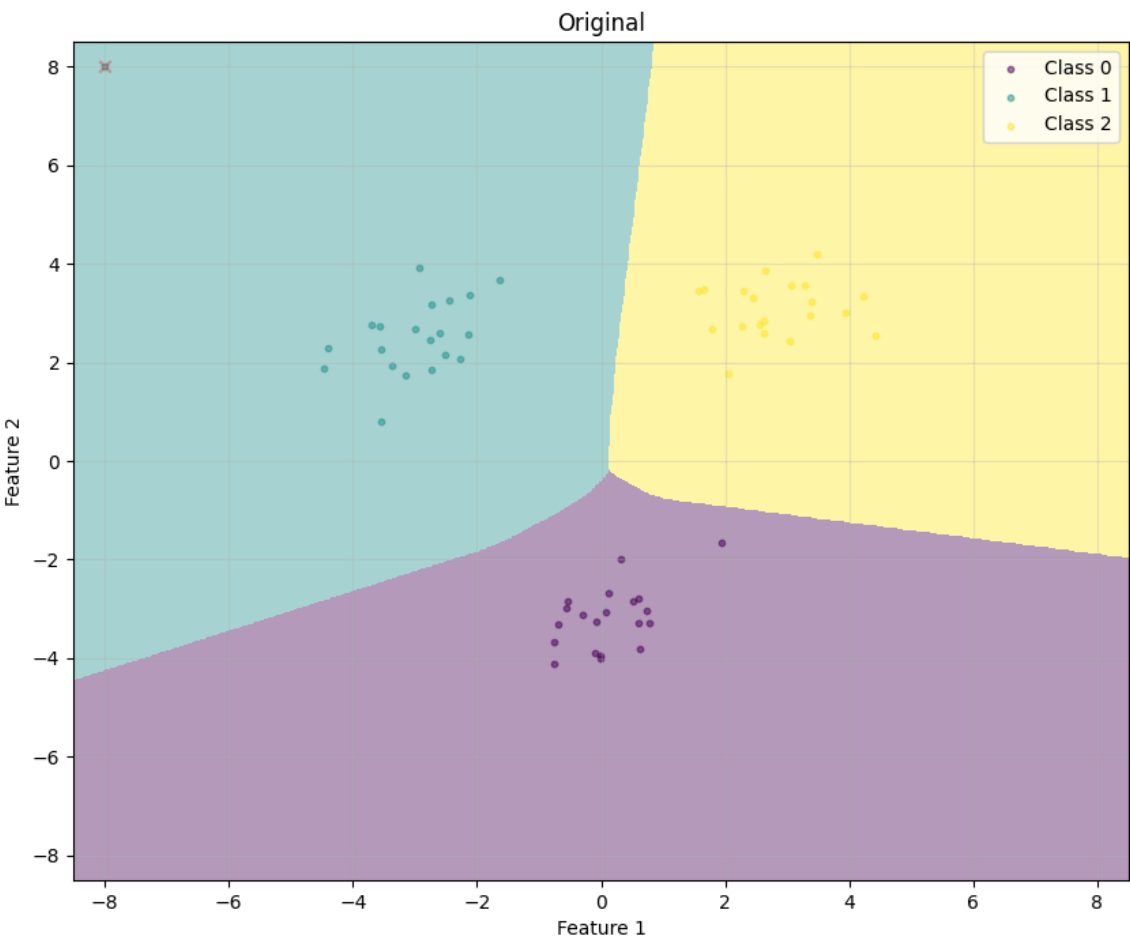




4. Rogue point far away from its centroid, but probably in the same decision boundary



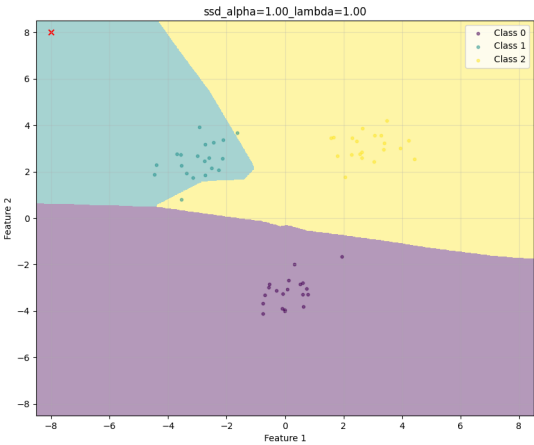
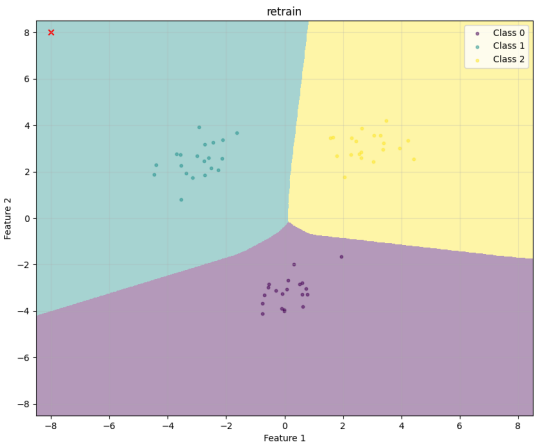
Original model for the dataset

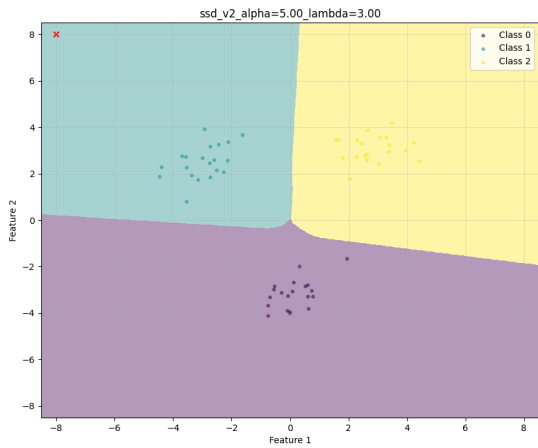
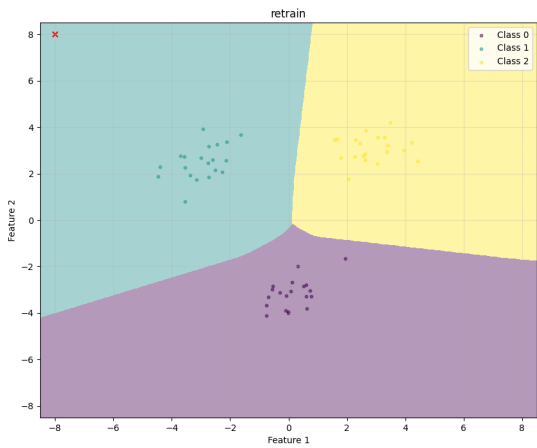
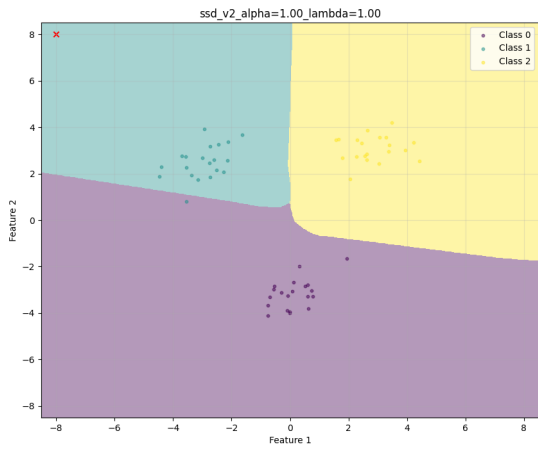
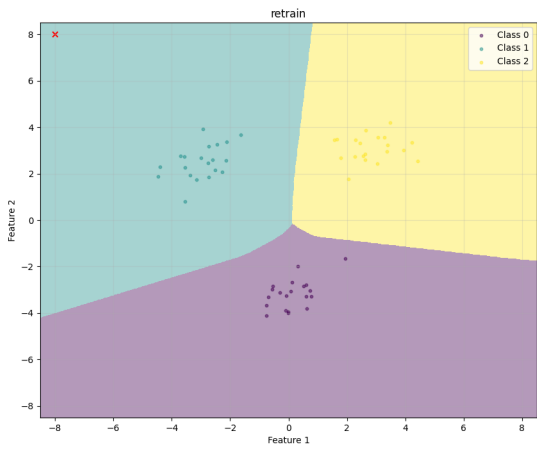
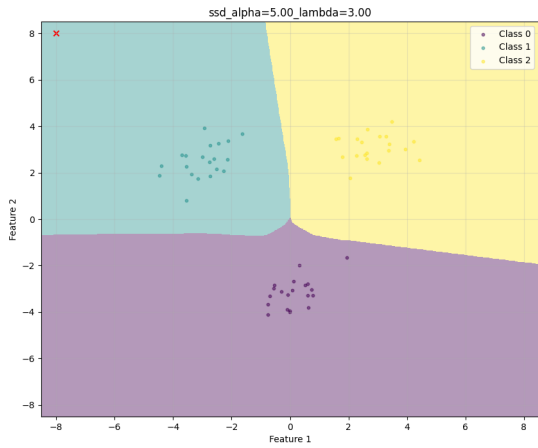
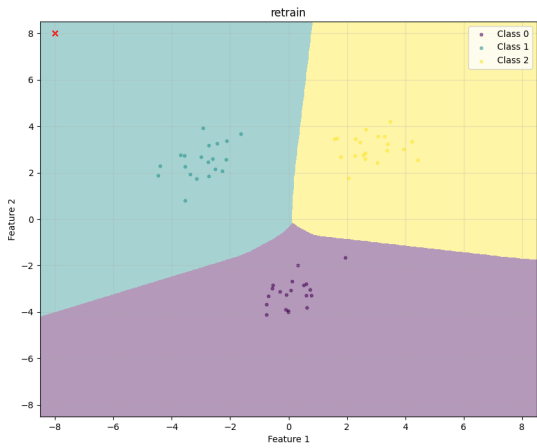


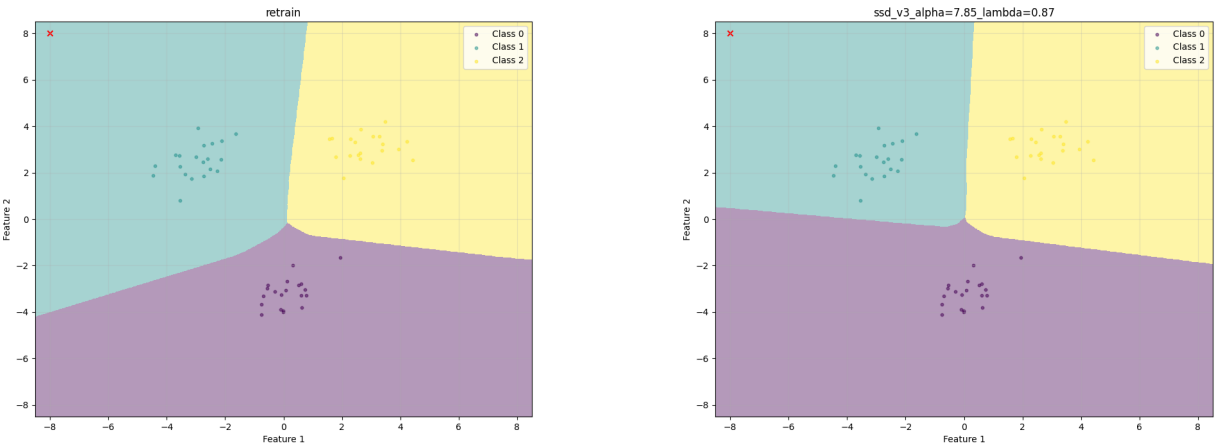
Findings

Unlearned

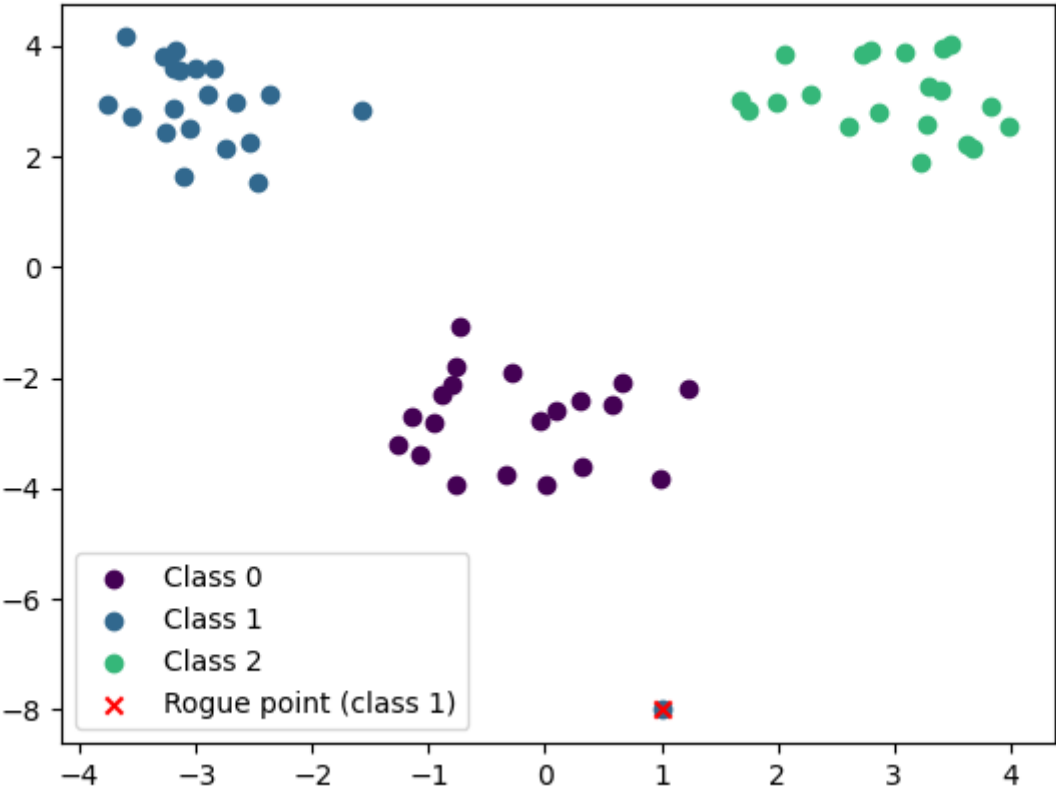
SSD



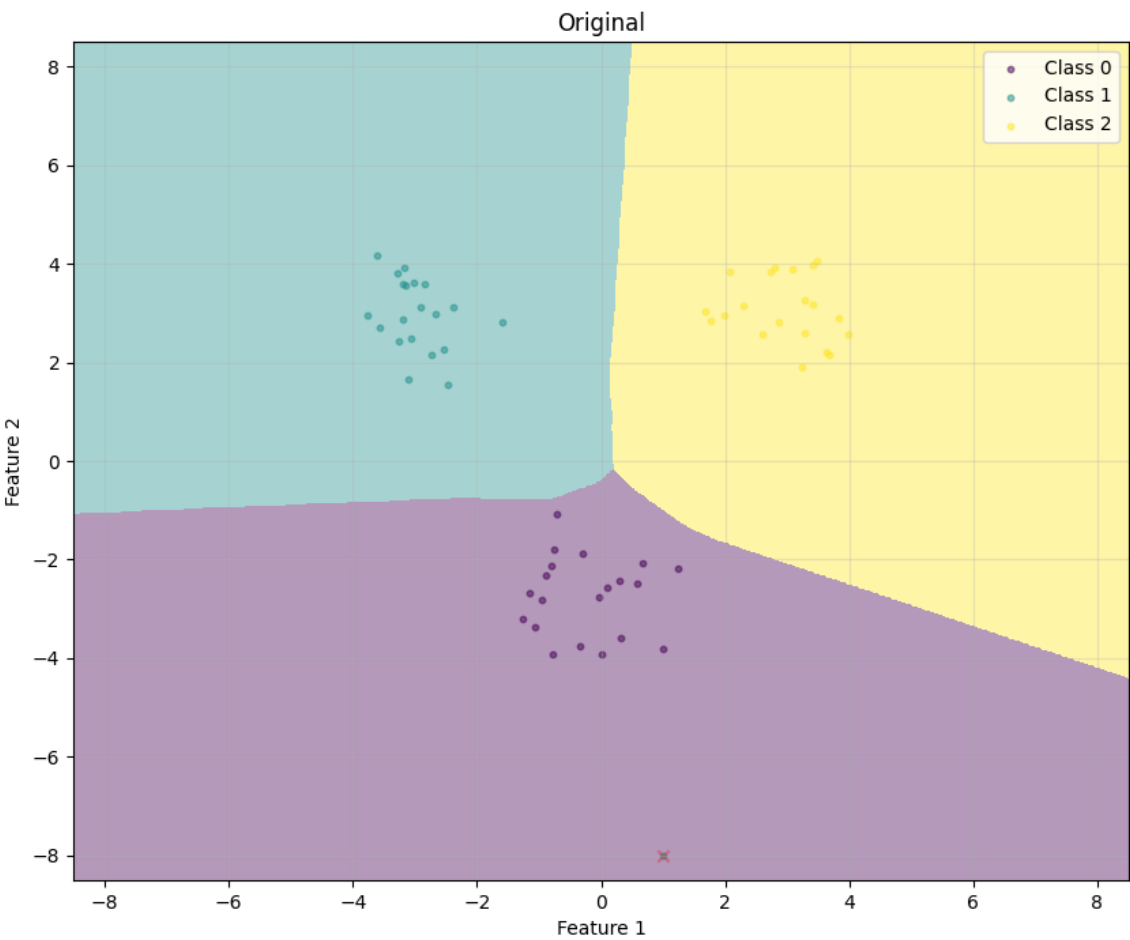




5. Rogue point far away from its centroid, but probably in the same decision boundary



Original model for the dataset



Findings

Unlearned

SSD

